Feature and UX assessment by foundry size

Foundry ERP (c-foundry-erp4) · 22 October 2026 · Companion to docs/AUDIT.md

Basis. This assessment comes from reading the code that decides how the system scales: process-depth presets, module licensing, QuickFlow, Quick QC, dashboards, roles, navigation, the responsive layout and the domain modules, plus the earlier ISO/ASTM audit. The app was not clicked through, so UX verdicts come from the views and controllers, not from user testing.

What the system is today

A **green-sand grey and ductile iron jobbing foundry ERP** with very good melt control and a clear idea of “light” versus “full” process. Defaults such as PKR, Jhelum river sand and molasses (sheera) in sand tests show it is built for South Asian iron foundries first.

Scale-related building blocks already in place:

Building block	Where	What it does for scale
Process-depth presets BASIC / FULL, 10 rules each editable	app/Support/ProcessDepth.php	Switches gates, holds, contract review, NCR threshold and lot grouping on or off
Module licensing and toggles	Licensing, config/modules.php	Hides HR, CAPA, Tooling, Purchasing and others when not needed
QuickFlow floor entry	Domains/QuickFlow	Type “moulds made” and “moulds poured”; real moulding-run and pour records are written behind it
Quick QC tally	QuickQcController	Good, rework and rejected counts per product over the oldest open lots
Bulk and pasted moulding-run entry	BulkRunEntry, BulkRunPaste	Spreadsheet-speed entry for clerks
Global search and scan box	ScanController, CodedRecords	Barcode or code lands on the record
4 dashboard layouts	dashboard/classic, columns, command, telemetry	Different density per user
9 seeded roles	config/permissions.php	Sales, production manager, melt and moulding supervisors, QC, stores and others
Responsive sidebar with a mobile mode	layouts/app.blade.php	Usable on a phone, though screens are not designed for one

Process coverage limit: iron_grades.family is constrained to iron families. Steel, aluminium/brass, investment casting and die casting are outside the model at every size.

The five foundry profiles

1. Micro jobbing foundry

5–30 people, a cupola or crucible, 1–5 t/day, owner-run, one shared PC or the owner's phone.

How it works today

- Turn on BASIC and license only the core.
- The owner books an order in QuickFlow (QuickOrderController).
- The floor clerk types “40 moulds made” and later “38 poured from heat 12”.
- Quick QC records good and bad counts.
- The pattern ownership screen tracks customer-owned patterns. The workflow is genuinely short.

Works

Few screens. Each order completes once its castings are poured. One lot per pour. No forced gates. Customer codes are generated automatically.

Missing

- **No billing or cash.** No invoice, payment or customer balance, which is the first thing a micro foundry owner wants. Quotations help, but the loop isn't closed.
- **Weight-based pricing and job costing.** Jobbing foundries quote and invoice per kg. Costing is an empty domain folder.
- **English only.** lang/ contains only en. Floor clerks at this size often need Urdu or another local language, and icons with numbers.
- **The full menu is still the full menu.** About 40 nav items under BASIC. Process depth does not hide screens; only module licensing does.
- **Accountability.** Login is off by default (FOUNDRY_REQUIRE_LOGIN=false) and entered_by / operator_name are free text. Acceptable here, but the wrong default for anyone bigger.

UX verdict

Close. It needs a “Micro” home showing only Orders, Floor entry, Quick QC and Dispatch/Invoice, with large touch targets.

Fit: 6 / 10

2. Small foundry

30–120 people, 1–2 induction furnaces, 5–25 t/day, hand or jolt-squeeze moulding, first ISO 9001 customers.

How it works today

- This is the system's sweet spot.
- Sales order, then contract review (optional), then production order.
- Moulding work lists with teams and stations, then the three-step moulding run screen.
- Melting plan, heat timeline with chemistry import and thermal cups, then pour.
- Lots move through the production queue (shakeout to final inspection), then the certificate page, then packing and dispatch.
- Sand lab, patterns with revisions and cavity map, gauges, HR and payroll are all available.

Works

- Per-rule process depth lets a small shop enable the thermal gate for ductile only, or require contract review without other holds.
- Heat plan with furnace eligibility.
- Bulk run paste.
- The payroll module is unusually complete for this size, which matters because piece-rate moulding teams are common.

Missing

- Purchase orders and receipts: Purchasing covers suppliers and incoming inspection only.
- Stock-based MRP (empty folder). Charge-cost or least-cost charge.
- Invoicing.
- Casting-level costing (kg poured, yield, rejects, labour from runs).
- A controlled certificate (see docs/AUDIT.md).
- Scrap reasons that come back from customers.

UX verdict

- Desktop screens are dense but good.
- The shop floor is the weak spot. Furnace and moulding entry need a **tablet mode**: numeric keypad, one record per screen, scan-first, and survival of drop-outs near furnaces (no PWA or offline queue).

- Role-based start pages would cut most of the navigation. A melt supervisor should land on today's heats, not the dashboard.

Fit: 8 / 10

3. Medium certified foundry

120–400 people, a semi- or fully automatic moulding line, 25–100 t/day, ISO 9001, export customers needing 3.1 certificates.

How it works today

- FULL preset.
- Enforced pour gate, first-iron and containment holds, completion after final inspection.
- NCR from any reject. CAPA module.
- Controlled documents, control plans and inspection plans, approvals, audit log.
- Gauge calibration blocks or warns.
- Lot grouping by MOULD for traceability.

Works

The quality backbone is real. The gates are enforced, not decorative.

Missing

- **Line-speed data entry.** An automatic line makes 100–400 moulds an hour. Run-by-run screens assume a human records each run, so this tier needs counts from the line PLC (no API) and lot grouping by shift or DAY, which STATUS lists as not done.
- Finite capacity scheduling across line, cores and furnaces. HeatPlanner covers melt only.
- Core shop records: definitions and boxes exist, but no core runs.
- NDT and post-cast heat treatment records.
- Signed, numbered 3.1 certificates.
- Defect Pareto by pattern, sand or heat.
- Named users on every record: free-text names fail an ISO audit trail at this size.
- Energy kWh/t reporting.

UX verdict

- Needs exception-first dashboards: “what is on hold, why, and who must act”. _attention.blade.php is a start.
- Needs approval inboxes per role.
- Needs fewer forms and more scan-and-confirm.
- Supervisors need a shift handover view.

Fit: 6.5 / 10

4. Large automotive / IATF foundry

400–1500 people, several lines, IATF 16949, PPAP, SPC, MES on the floor.

How it works today

It would run as in tier 3 with the IATF requirement profile attached to customers. But that profile only *declares* the PPAP level, FMEA format and 8D. Nothing records or produces them.

Works

Everything in tier 3, plus customer requirement profiles that capture what each customer demands.

Missing

- PPAP packages, PFMEA linked to the control plan, MSA (gauge R&R), SPC with Cpk/Ppk and reaction plans.

- Layered process audits (the lpa_required flag exists, but there are no records).
- Customer scorecards, EDI/ASN, serialisation or part marking.
- An MES/SCADA ingest API and an ISA-95 hierarchy.
- Downtime and OEE per line.
- Andon.
- Recall where-used queries.

UX verdict

This tier expects live line boards on big screens, SPC charts at the station, and operator sign-on by badge. None of those patterns exist yet. The current UI is built for record entry, not monitoring.

Fit: 3.5 / 10

5. Multi-plant group

Several sites, mixed processes (iron, steel, aluminium, investment, die casting), group finance.

How it works today

One installation per plant, with no link between them.

Works

Each plant gets a capable single-site system; HR, documents and quality modules carry over unchanged.

Missing

- Site or plant scoping on records, and inter-plant transfers.
- Group consolidation, multi-currency (currency is per material lot only), and a finance or GL connector.
- Non-iron alloy families, investment-casting (wax, shell, slurry) and die-casting (die, shot profile, cycle) models.
- SSO, and an API for the group BI.

UX verdict

Would need a site switcher, cross-site search and group dashboards.

Fit: 2 / 10. It would be a deliberate platform step, not an increment.

Coverage matrix

Yes = there · Partial = partly there · No = missing · – = not needed at that size

Capability	Micro	Small	Medium	Large / IATF	Multi-plant
Quick order, floor counts, QC tally	Yes	Yes	–	–	–
Full order, mould, melt, pour, lot	–	Yes	Yes	Partial (no line data)	Partial
Melt control (chemistry, thermal, gate)	–	Yes	Yes	Yes	Partial (iron only)
Invoice / receivables / costing	No	No	No	No	No
Purchase orders / MRP	–	No	No	No	No
Certificates (controlled 3.1)	–	Partial	Partial	Partial	Partial
NCR / CAPA / documents / gauges	–	Yes	Yes	Partial (no 8D)	Yes
NDT, casting HT, core shop	–	No	No	No	No
PPAP / FMEA / MSA / SPC	–	–	Partial (control plans)	No	No
Machine / device ingest API	–	–	No	No	No
HR and payroll	Partial	Yes	Yes	Yes	Partial (one site)
Local language	No	No	No	No	No

Capability	Micro	Small	Medium	Large / IATF	Multi-plant
Tablet / offline shop-floor mode	No	No	No	No	No
Named-user accountability	–	Partial	No (free text)	No	No
Site hierarchy	–	–	–	No	No

Architectural and UX recommendations

These fit the existing rule that each process step is a per-rule setting and small foundries get few screens.

1. **Grow BASIC/FULL into tiered presets** (Micro, Small, Certified, Automotive) that set both rules *and* screen visibility. Every rule stays editable on its own. Today process depth changes behaviour but not navigation, so a Micro foundry still sees about 40 menu items.
2. **Role-based start pages.** Each role lands on its own work queue: melt supervisor on today's heats, moulder on floor entry, QC on the lot queue, sales on orders. This saves more navigation than any menu redesign.
3. **Shop-floor tablet mode.** A separate layout for floor screens: large targets, numeric keypad, scan-first, one action per screen, and an offline entry queue (PWA) for poor Wi-Fi near furnaces.
4. **Operator identity without friction.** A PIN or badge sign-on per tablet that fills entered_by from a real employee record. Keep free text available for Micro.
5. **Localisation.** Move UI strings to lang/ and add Urdu, including RTL. This is the cheapest way to win the Micro and Small tiers already being targeted.
6. **Close the money loop for Micro and Small.** Per-kg invoicing from dispatch, receivables and simple job costing (kg poured, yield, rejects, labour). These matter more than IATF features for most of the near-term market.
7. **Ingest API and site scoping before chasing tiers 4–5.** Both are cross-cutting schema changes, and they get more expensive the longer they wait.

Suggested order: 6 and 5 (win the Micro and Small market), then 1, 2 and 3 (UX that scales), then audit items A and B (certificates, NDT) for Medium, then 4 and 7, then IATF features.